

CH. 10

EXPENDITURE CYCLE

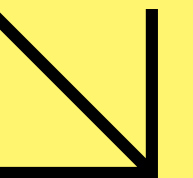


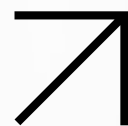
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- Expenditure Cycle Activities and Technologies
- Expenditure Cycle Audit Objectives, Controls, and Tests of Control
- Substantive Tests of Expenditure Cycle Accounts

01

EXPENDITURE CYCLE ACTIVITES AND TECHNOLOGIES



01

ALTERNATIVE INFORMATION TECHNOLOGIES



THREE ALTERNATIVES

1. Purchases and Cash Disbursement System – Batch Processing
2. Modern System – Real Time (Reengineered)
3. Modern Payroll System

01

PURCHASE & CASH DISBUREMENT SYSTEM



BATCH PROCESSING

Key Features

- Batch processing with sequential files.
- Early legacy system, single-user, no data sharing

01

FOUR (4) STEPS UNDER PURCHASES



BATCH PROCESSING

1. Inventory Control in Data Processing Department
2. Create Open Purchase Order Record
3. Update Inventory Subsidiary File
4. Validation of Vendors

01

INVENTORY CONTROL IN DATA PROCESSING DEPARTMENT



STEP 1

- System checks inventory subsidiary file for reorder points.
- Creates record in open requisition file if needed.
- Sorts requisitions by vendor, consolidates items.
- Retrieves vendor info, produces purchase requisition documents.
- Copies sent to purchasing and accounts payable

PURCHASING DEPARTMENT

- Prepares a five-part purchase order.
- Copies distributed to vendors, AP, receiving, data processing, and purchase file.

(3) DIFFERENT ALTERNATIVE INVENTORY ORDERING PROCEDURES

- **Alt 1:** PO prepared → reviewed & signed → mailed to vendors.
- **Alt 2:** PO distributed directly to vendors & users; transaction list for review.
- **Alt 3:** Automated ordering via telecommunications link (EDI).

01

CREATE OPEN PURCHASE ORDER RECORD



STEP 2

- Transfers requisition records to closed files.

RECEIVING DEPARTMENT

- Clerk prepares receiving report when goods arrive.
- Copies sent to purchasing, AP, and data processing.

01

UPDATE INVENTORY SUBSIDIARY FILE



STEP 3

- Batch totals calculated for GL update.
- Records closed in purchase order file.

ACCOUNTS PAYABLE

- Clerk reconciles supplier invoice with supporting documents.

01

VALIDATION OF VENDORS



STEP 4

- Batch program validates voucher records against vendor file.
- Adds to voucher register/open AP subsidiary file.
- Prepares batch totals for posting to AP control account.

01

(3) STEPS UNDER CASH DISBURSEMENT



BATCH PROCESSING

- Step 5:** Scan for items due & print checks.
- Step 6:** Reconcile checks with transaction listing, submit for signing.
- Step 7:** AP Clerk matches checks with vouchers, closes them, files documents.

01

MODERN SYSTEM- REALTIME



REENGINEERED

Key Features

- Real-time procedures, direct-access files.
- Eliminates routine manual procedures.
- Uses electronic communications between departments

01

REENGINEERED

MODERN SYSTEM-
REALTIME

AUTOMATED DATA PROCESSING

- a. Inventory file searched for reorder points.
- b. Record entered in purchase requisition file.
- c. Requisitions consolidated by vendor.
- d. Vendor info retrieved.
- e. Purchase orders prepared & sent (or via EDI).
- f. Transaction added to open PO file.
- g. Transaction listing sent to purchasing for review.

RECEIVING DEPARTMENT

- Clerk enters PO number from packing slip.
- System prompts for quantities received.

MODERN SYSTEM- REALTIME

01 REENGINEERED

DAILY AP PROCEDURES

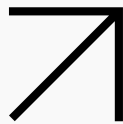
1. Print, sign, and distribute checks (or electronic payments via EDI).
2. Record payments in check register file.
3. Transfer paid items to closed AP file.
4. Update GL accounts payable & cash.
5. Reports transmitted to AP & cash disbursement departments.

CONTROL IMPLICATIONS

- Risks: segregation of duties weakened, unauthorized access.
- Solutions: detailed transaction listings, access controls, monitoring reports.

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AUTOMATED VS REENGINEERED SYSTEM



AUTOMATED SYSTEM

A process where the old system is still used, but with the help of technology to perform task with minimal human involvement.

ADVANTAGES



- 1.Improved Inventory Control
- 2.Better Cash Management
- 3.Less Time Lag
- 4.Better Purchasing Time Management
- 5.Reduction of Paper Document

CONTROL IMPLICATIONS



- 1.Errors in the system can disrupt inventory levels.
- 2.Automation increases the risk of unauthorized payments.

01

AUTOMATED VS REENGINEERED SYSTEM



REENGINEERED SYSTEM

A redesigned process that uses technology to simplify, automate, and improve business operations.

ADVANTAGES



- 1. Real-time processing reduces delays in recordkeeping.
- 2. Automation removes routine manual procedures.
- 3. Electronic communication and storage reduce paper use.

CONTROL IMPLICATIONS



- 1. Segregation of Duties
- 2. Accounting Records and Access Controls

01

OVERVIEW OF PAYROLL PROCEDURES ↘

PAYROLL

The compensation a company pays its employees for work over a specific period or on a set date.

PAYROLL PROCESSING

A special expenditure system where a company pays employees for labor instead of purchasing goods, requiring unique procedures and controls.

The nature of payroll processing creates the need for **specialized procedures** for the following reasons:

- 1. General expenditure procedures that apply to all vendors will not apply to employees.
- 2. Writing checks to employees requires special controls.
- 3. General expenditure procedures are designed to accomodate a relatively smooth flow of transactoins.

The use of **Batch Processing** and **Sequential Files** are suitable because payroll is processed periodically (weekly and monthly), not continuously.

- 1. **Time Cards**
Records employee working hours to determine payroll amounts
- 2. **Job Tickets**
Record time spent on specific jobs to assign labor costs to production (Work in Process)

Control Implications:
Improves accuracy and reducess payroll errors but does not greatly increase overall operational efficiency.

- 3.**Modern Payroll System**
A computerized, integrated system that automates employee payment processing.

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REENGINEERING THE PAYROLL SYSTEM



Payroll is often integrated with Human Resource Management (HRM) system in medium and large organizations, allowing real-time access to employee records and immediate updates of any changes in employee status.

DIFFERENCE FROM SIMPLE AUTOMATED SYSTEM

- 1. Operations departments transmit transactions to data processing via terminals.
- 2. Direct access files are used for data storage.
- 3. Many processes are now performed in real time.

KEY OPERATING FEATURES (REAL-TIME)

Personnel
Updates employee records in real time using terminals.

Cost Accounting
Records job cost data daily or in real time to generate the labor usage file.

Timekeeping
Updates the current attendance file after receiving approved timecards weekly.

DATA PROCESSING



At the end of the work period, the following tasks are performed in a batch process:

01

1. Labor costs are distributed to various work-in-process, overhead, and expense accounts.
2. An online labor distribution summary file is created.
3. An online payroll register is created from the attendance file and the employee file.
4. The employee records file is updated.
5. Payroll checks are prepared and signed.
6. The disbursement voucher file is updated and a check is prepared for the funds transfer to the payroll imprest account.
7. At the end of processing, the system retrieves the labor distribution summary file and the disbursements voucher file and updates the general ledger file.

02

EXPENDITURE CYCLE

**AUDIT OBJECTIVES, CONTROLS, AND
TEST OF CONTROLS**



02

MANAGEMENT ASSERTIONS	AUDIT OBJECTIVES
EXISTENCE OR OCCURENCE	Verify that recorded liabilities, and transactions actually exist and occurred.
COMPLETENESS	Ensure all transactions and balances that should be recorded are included
ACCURACY	Ensure financial data is recorded correctly
RIGHTS AND OBLIGATIONS	To confirm that recorded liabilities are valid, legally enforceable obligations of the entity and properly recognized in the financial statements.
VALUATION OR ALLOCATION	To ensure expenses and liabilities in the expenditure cycle are recorded at correct amounts and allocated to the proper accounting period.
PRESENTATION AND DISCLOSURE	To ensure financial information is clearly presented and fully disclosed in the financial statements.

APPLICATION CONTROL FRAMEWORK

- Input Controls
- Process Controls
- Output Controls

DATA VALIDATION CONTROLS

- Data validation controls ensure that input data is accurate, complete, and reliable.
- They help prevent errors before data is processed or stored.

TYPES OF DATA VALIDATION CONTROLS

- **Missing Data Check**→ ensures required fields are not left blank
- **Numeric-Alphabetic Check**→ ensures correct data type is entered
- **Limit Check**→ ensures value does not exceed a fixed limit
- **Range Check**→ ensures values falls within an allowed range
- **Validity Check**→ ensures data matches predefined rules or lists
- **Check Digit**→ detects errors in codes like account numbers or barcodes



02

TESTING VALIDATION CONTROLS

- Refers to the process of checking whether data validation controls are working correctly.
- It is the verification stage where we test if the validation rules actually prevent errors as intended.

TWO COMMON TESTING APPROACH

1. Integrated Test Facility
2. Test Data Approach

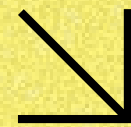
BATCH CONTROLS

- Batch controls are used when processing large volumes of transactions in groups
- They ensure data is complete, accurate, and free from duplication or loss.



02

PURCHASE AUTHORIZATION CONTROLS



PURCHASE AUTHORIZATION

The internal control process that ensures inventory is reordered only when necessary and in authorized quantities.

It actually occurs in the revenue cycle when goods are sold to customers.

To determine if the inventory needs to be reordered

1. The system compares the quantity-on-hand with the reorder point.
2. If the reorder point is reached, the system creates a purchase requisition for the standard order quantity.
3. The item is flagged "On Order" to prevent subsequent orders from being placed.

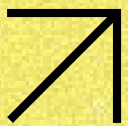
Purpose: to prevent subsequent orders from being placed

Expenditure Cycle

- The purchase requisition file is sorted by vendor number;
- Purchases orders are then prepared

02

TESTING PURCHASE AUTHORIZATION CONTROLS



AUDIT OBJECTIVE

To prevent overinvestment in inventory caused by unnecessary or incorrect orders.

Key Assertions:
Accuracy and Valuation

AUDITOR CONCERNS

- Verifying that the correct order quantity is used when a requisition is created.
- Ensuring the inventory record is properly flagged as “On Order.”
- Failure: Will result in overinvestment in inventory due to multiple orders being placed for the same time.

AUDIT PRECEDURE (CAATTS)

Involves creating test inventory records and sales transactions that reduce inventory items below their reorder point to test the system’s response.

02

EMPLOYEE
AUTHORIZATION



PERSONNEL ACTIONS FORMS

Documents prepared by the personnel department to effect changes in hourly pay rates, payroll deductions, and job classification

PURPOSE OF AUTHORIZATION

To identify employees who are authorized to receive a paycheck and prevent errors

WHAT KINDS OF PAYROLL FRAUD CAN POSSIBLY OCCUR?

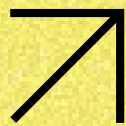
- Submitting timecards for individuals not employed by the firms.
- Creating checks for “ghost,, employees (non-existent workers)

FRAUD PREVENTION

A desgated indiviual in the payroll department should compare timecards with the authorized employee list from personnel.

02

TESTING EMPLOYEE AUTHORIZATION PROCEDURES



AUDIT PROCEDURE

The auditor assesses programmed procedures that validate employee authenticity using **Test Data** or an **Integrated Test Facility (ITF)**

- Needs to determine that effective procedure exist in the personnel department to identify current employees,
- Communicate their status completely,
- Correctly to the payroll function,
- Monitor adherence to employee authorization.

ASSURANCE REQUIREMENTS

- Access to the authorized employee file must be **password controlled**.
- Additions to and deletions from the file are restricted to authorized personnel.
- Employee records stored on the file must be **encrypted**.

Key Assertions:

Existence, Accuracy, Valuation, and Rights and Obligation

02

PROCESS CONTROLS

Includes file update controls, access controls, and physical controls.

FILE UPDATE CONTROLS

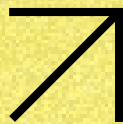
Safeguards that ensure each system run processes a batch correctly and completely

VALIDATION METHOD

Key fields, such as **invoice ammounts** and **record counts**, are compared to values in the **transmittal record**.



CONTROL TECHNIQUE



SEQUENCE CHECK CONTROL

A process where the batch must be re-sorted int he order of the master file used in each run.

LIABILITY VALIDATION CONTROL

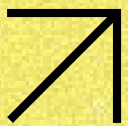
The process of reconciling supporting documents (Purchase Order, Receiving Report, and Supplier’s Invoice) before an account payable is recognized.

VALID VENDOR FILE

A list of vendors with whom the organization normally does business; non-matching records are rejected and investigated by management.

02

TESTING FILE UPDATE CONTROLS



TESTING SEQUENCE CHECK

performed using either ITF or the test data approach.

TESTING LIABILITY VALIDATION

Creating test purchase orders, receiving reports, and invoices to verify if the system’s decision rules are correctly applied.

TESTING VALID VENDOR CONTROL

Creating a reference file of valid vendors and a file of supplier invoices to be paid to test refaction logic.

RISKS OF FAILURE

Transaction not being processed, payments approved for unauthorized recipients, or posting to the wrong supplier’s account.

02

ACCESS CONTROLS

Procedures designed to prevent and detect unauthorized and illegal access to firm assets like inventory and cash.

Traditional Techniques

Warehouse security (fences/alarms), daily bank deposits, and locking cash drawers/safes.

Risks of Weak Access Control

- Unauthorized individuals adding their own accounts to the **AP subsidiary ledger**.
- Illegal triggering of paychecks via access to employee attendance cards.
- Theft of physical inventory paired with digital record adjustment to cover the crime.

TESTING ACCESS CONTROLS

Access control is the **heart of accounting information integrity**.

The state where data is accurate, complete, and protected from unauthorized deletion or falsification.

Audit Focus

Access to operating systems, networks, and databases that interact with applications

Testing Procedures

- Assess password controls and restriction of access to authorized managers.
- Verify the use of data encryption to prevent file contents from being read or changed

02

PHYSICAL CONTROLS

Include manual activities and human actions to initiate computer procedures to safeguard the assets of the organization.

Purchase System

Includes Segregation of inventory control from the warehouse, segregation of the general ledger and AP from cash disbursements, and supervision of receiving department to prevent theft of assets.

Payroll System

Includes verificaiton of timecards by supervisors, use of an independent paymaster, and a payroll imprest account at the bank.

Testing

Reviewing segregation of duties to test all management assertions.

TESTING PHYSICAL CONTROLS

Inadequate segregation of duties and lack of supervision can result in fraud and errors that can cause financial statements to be materially misstated.

Granting inappropriate access privileges is often due to assigning incompatible duties to an individual.

02

OUTPUT CONTROLS

Designed to ensure information is not lost, misdirected, or corrupted.

Accounts Payable Change Report

A summary report showing overall changes to AP.

Transaction Logs

Serves as journal for every transaction successfully processed.

Transaction Listing

Hard copies of successful transactions sent to users to facilitate reconciliation.

Log of Automatic Transactions

Maintains an audit trail of transactions triggered internally by the system.

Unique Transaction Identifiers

Using a unique number (like a PO number) to trace a particular transaction through the database.

Error Listing

A list of all error records sent to users for error correction and resubmission.

TESTING OUTPUT CONTROLS

Evidence gathered relates to the completeness and accuracy of assertions.

This involves reviewing summary reports for accuracy, completeness, timeliness, and relevance to the decision that they are intended.

02

PURCHASE SYSTEM CONTROLS



SEGREGATION OF INVENTORY CONTROL FROM THE WAREHOUSE

Warehouse clerks responsible for asset custody should not be given responsibility for maintaining inventory records.

SEGREGATION OF THE GENERAL LEDGER AND ACCOUNTS PAYABLE FROM CASH DISBURSEMENTS

An individual with the combined responsibilities of writing checks, posting to the cash account, and maintaining accounts payable could perpetrate fraud against the firm.

SUPERVISION OF RECEIVING DEPARTMENT

Close supervision here reduces the chances of two types of exposure: failure to properly inspect the assets and the theft of assets.

a. Inspection of assets

To obtain the information on quantities.

b. Theft of assets

Incoming inventories are exposed to theft until they are securely placed in the warehouse

RECONCILIATION OF SUPPORTING DOCUMENTS

Each document contains unique facts about the purchase transaction, which the accounts payable clerk needs to verify before the obligation is recognized.

a. Purchase Order

shows that the purchasing agent ordered only the needed inventories from a valid vendor.

b. Receiving Report

evidence of the physical receipt of the goods.

c. Supplier's invoice

provides the financial information needed to record this obligation as an AP.

SUBSTANTIVE TEST

03

- Expenditure Cycle Risks and Audit Concerns
- Understanding Data
- File Structure for the Expenditure Cycle
- Testing the Accuracy and Completeness Assertions
- Review Disbursement Vouchers for Unusual Transactions and Account Balances
- Reviewing for Accurate Invoice Prices
- Testing the Completeness, Existence, and Rights and Obligation Assertions

FILE STRUCTURE

03

1. Inventory File
2. Purchase Order File
3. Purchase Order Line-Item File
4. Receiving Report File
5. Disbursement Voucher File and Check Register File
6. File Preparation Procedures

TESTING ASSERTIONS

ACCURACY AND COMPLETENESS

03

In the case of the expenditure cycle, an analytical review can provide the auditor with an overall perspective on trends in accounts payable and related expenses.

Analytical procedures may indicate trends, even in adequately controlled organizations, that lead the auditor to extend substantive tests.

On the other hand, they can provide assurance that transactions and accounts are reasonably stated and complete and may thus permit the auditor to reduce substantive testing.

REVIEW DISBURSEMENT VOUCHERS

**FOR UNUSUAL TRANSACTIONS AND
ACCOUNT BALANCES**

03

SCANNING

Usual audit procedure for identifying whether data files contain unusual transactions and account balances.

ACLS STRATIFY AND CLASSIFY FEATURES

Use to identify various characteristics and anomalies associated with accounts payable procedures.

Stratify function groups financial data into specific strata

Classify function forms strata based on non-financial and non-numeric data

REVIEWING FOR

ACCURATE INVOICE PRICES

03

Comparing prices on supplier invoices to original purchase order prices provides evidence for testing the management assertion of accuracy.

Significant discrepancies between expected prices and the prices actually charged may be due to clerical errors, failure to review supporting documents before authorizing payment or accounts payable personnel exceeding their authority in dealing with price discrepancies.

USING AUDIT COMMAND LANGUAGE (ACL)

Allows the auditor to compare prices charged on every invoice in the file for a period under review.

Using ACL's Expression Builder feature, the auditor can easily create a filter that will select only records where PO amount and amount are not equal. The resulting file will thus contain only records with price discrepancies.

TESTING THE

**COMPLETENESS, EXISTENCE, AND RIGHTS AND
OBLIGATION ASSERTIONS**

03

- Searching for Unrecorded Liabilities
- Searching for Unauthorized Disbursement Vouchers
- Review for Multiple Checks to Vendors
- Auditing Payroll and Related Accounts

THE END